

1. Administrative & Study Identification

Full Study Title: An open-label, multi-center, phase I/II study to assess the safety, tolerability and efficacy of DFT383 in pediatric participants with nephropathic cystinosis, followed by a long-term extension phase. **Short Title/Acronym:** CYStem Trial. **Protocol Number:** DFT383-001 (Internal Novartis Identifier). **NCT Number:** NCT06910813. **Sponsor Name:** Novartis. **Investigational Product:** DFT383 (Cellular Gene Therapy). **Phase of Development:** Phase I/II. **Medical Monitor:** Novartis Clinical Operations Lead.

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, and efficacy of the cellular gene therapy DFT383 in pediatric participants aged 2 to 5 years with nephropathic cystinosis. **Secondary Objectives:** To evaluate the long-term preservation of kidney function and compare outcomes against a concurrent reference arm of patients receiving the standard of care.

2.2 Background & Rationale To address the significant disease burden and demanding 24/7 treatment protocol of standard of care (oral cysteamine) for nephropathic cystinosis, the study evaluates a novel cellular gene therapy (DFT383) intended to correct the underlying genetic defect, reduce the buildup of cystine in cells, and prevent long-term kidney and organ damage in pediatric patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional. **Control Type:** Concurrent Reference Arm (Standard of Care - Cohort 0). **Blinding:** Open-label. **Randomization:** Non-randomized (Parallel cohorts).

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 30 participants (15 in Cohort 1 and 15 in Cohort 0). **Number of Sites:** Multi-center (Global/USA). **Estimated Study Start:** 2024-2025. **Estimated Primary Completion:** Core Phase of up to 32 months (Cohort 1) and up to 24 months (Cohort 0).

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Pediatric participants aged 2 to 5 years (inclusive). 2. Confirmed diagnosis of nephropathic

cystinosis. 3. Has been receiving oral cysteamine therapy (Standard of Care) for at least 6 months.

4.2 Exclusion Criteria 1. Unwilling or unable to comply with strict study procedures or the long-term follow-up requirements. 2. Any concurrent medical condition or abnormality that would compromise participant safety, including a severe or uncontrolled medical disorder. 3. Prior receipt of an investigational gene or cellular therapy, or a prior or planned bone marrow or stem cell transplantation. 4. A history of kidney transplantation. 5. History of malignancy. 6. Major surgery within 90 days.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Cohort 1:** Cellular gene therapy (DFT383) administered in a staggered approach across 3 sub-cohorts (1A, 1B, and 1C). - **Cohort 0:** No study treatment; will continue with standard of care (SoC). **Route of Administration:** Parenteral/Intravenous for DFT383; Oral for SoC. **Duration of Treatment:** Cohort 1 involves a Core Phase of up to 32 months, followed by a long-term extension phase spanning up to 13 years. Cohort 0 participants will be followed for up to 24 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care (cysteamine) is permitted and serves as the required baseline therapy for the Cohort 0 reference arm. **Prohibited:** Other investigational drugs or novel cellular/gene therapies for the duration of the core phase.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Baseline Labs, Gene Therapy Eligibility Assessment
Baseline	Day 0	Enrollment, Dosing of DFT383 (Cohort 1) or Continuation of SoC (Cohort 0)
Interim	Core Phase	Safety monitoring, Efficacy Assessments (Urine protein/creatinine, etc.)
End of Study	Month 32 (Core)	Transition to the 13-year long-term Extension Phase (for Cohort 1)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Improvement of proximal tubular function defined by a 2-fold change from Baseline in at least 4 out of 5 Renal Fanconi Syndrome (RFS) parameters (urine protein to creatinine ratio, urine amino acids, urinary glucose to creatinine ratio, tubular maximum reabsorption of Phosphate/Glomerular Filtration Rate, and urine retinol-binding protein/creatinine ratio). Normalization of a parameter also constitutes

improvement. **Measurement Method:** Central Laboratory Analysis of blood and urine samples.

7.2 Secondary & Exploratory Endpoints - Overall safety and tolerability of DFT383 in a pediatric population. - Long-term preservation of kidney function (eGFR) and reduction of cellular cystine accumulation.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via site visits, lab assessments, and caregiver reports. **Serious Adverse Events (SAEs):** Standard protocol-mandated 24-hour reporting timelines. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) utilized to review safety data iteratively, given the staggered dosing approach (Cohorts 1A, 1B, 1C) and pediatric gene therapy risks.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy; Safety Set for all participants receiving treatment. **Sample Size Justification:** Approximately 30 participants (15 per cohort) is deemed statistically and clinically appropriate to provide concurrent, descriptive comparison data in this rare disease setting. **Handling of Missing Data:** Standard multiple imputation (MI) or mixed models for repeated measures (MMRM).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure data integrity and protocol adherence. **Audit Trail:** Electronic Case Report Forms (eCRF) utilized with strict audit trails, database query resolution, and locked data review procedures.

11. Study Identifiers & Governance

Primary ID: CYStem / DFT383 **Registry Number:** NCT06910813 **Sponsor/Lead Organization:** Novartis **Study Title:** CYStem Trial: DFT383 in Pediatric Participants With Nephropathic Cystinosis **Official Regulatory Title:** An open-label, multi-center, phase I/II study to assess the safety, tolerability and efficacy of DFT383 in pediatric participants with nephropathic cystinosis, followed by a long-term extension phase.

12. Clinical Framework (Cystinosis Specific)

Primary Condition: Nephropathic Cystinosis **Diagnosis Threshold:** Confirmed clinical/genetic diagnosis of Nephropathic Cystinosis. **Medical Methodology:** Cellular Gene Therapy (DFT383) vs. Standard of Care (Cysteamine). **Optimization Target:** Improvement of proximal tubular function and preservation of the kidneys.

13. Study Procedures & Methodology

Management Pathway: Targeted gene therapy delivery vs. standard oral cysteamine. **Education Protocol (HCP):** Gene therapy administration, intensive safety monitoring, and RFS parameter tracking. **Education Protocol (Patient/Caregiver):** Informed consent counseling on the investigational nature of DFT383 and the necessity of up to 13 years of long-term follow-up. **Quality Audit Mechanism:** Independent safety data reviews between staggered sub-cohort (1A, 1B, 1C) dose escalations.

14. Observation & Follow-Up Schedule

Total Duration: Up to 32 months (Core Phase) + up to 13 years (Long-term Extension Phase). **Onsite Visit Cadence:** Intensive short-term monitoring for safety post-infusion, transitioning to scheduled follow-ups over the 32-month core period. **Remote Monitoring Frequency:** As per study protocol telemedicine check-ins. **Checklist Review Interval:** Regular laboratory surveillance of the 5 key RFS parameters and cellular cystine levels.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				